BCA 5TH SEMESTER • MACHINE LEARNING • UNIT IV

Multivariate Methods & Dimensionality Reduction

Factor Analysis, Multidimensional Scaling (MDS), and Linear Discriminant Analysis (LDA)

Primary Reference: Ethem Alpaydin, *Introduction to Machine Learning* (MIT Press / PHI)

CURRICULUM ROADMAP

Unit IV: Three Pillars of Multivariate Reduction

Factor Analysis (FA)

Uncovers hidden latent causes behind observed correlated variables.

Multidimensional Scaling (MDS)

Preserves pairwise proximity distances in a low-dimensional map.

Linear Discriminant Analysis (LDA)

Finds projection vectors maximizing between-class separability.

TOPIC 1 • FACTOR ANALYSIS

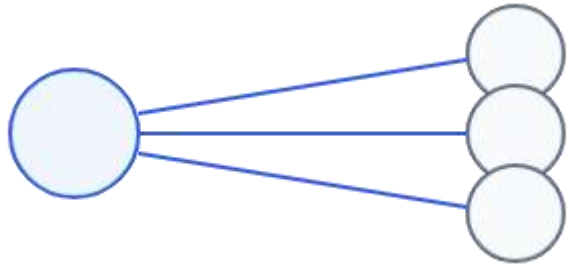
Core Intuition of Factor Analysis

High-dimensional datasets often exhibit high cross-correlation because unmeasured hidden forces generate them.

Unobservable Causes: Latent factors cannot be measured directly.

Data Compression: Replaces *d* observable variables with *k* latent factors (*k* << *d*).

Generative Nature: Models observed data as linear combinations of factors.



Latent Factor *z* drives multiple observed variables x_1, x_2, x_3

MOTIVATION • FACTOR ANALYSIS

Why is Factor Analysis Essential in Machine Learning?

Redundancy Elimination

Removes multi-collinearity among highly correlated features.

Latent Discovery

Uncovers psychological traits, market segments, or biological signals.

Model Regularization

Mitigates the curse of dimensionality for classifiers.

CONCEPTUAL COMPARISON

Observed Variables vs. Latent Factors

Observed Variables (*x*)

Measurement: Directly recorded and quantified.

Examples: Math test score, Physics score, Chemistry score.

Noise Status: Directly corrupted by sensor or grading noise.

Latent Factors (*z*)

Inference: Statistically deduced from covariance.

Examples: Quantitative aptitude, Analytical intelligence.

Standardization: Assumed zero mean and unit variance.

MATHEMATICAL FOUNDATION • ALPAYDIN FORMULATION

The Generative Factor Analysis Model

$$x - \mu = V z + \varepsilon$$

Generative linear model mapping *k* latent factors to *d* observable variables (*k* < *d*)

$z \sim N(0, I)$

Independent latent factors.

$V \in \mathbb{R}^{d \times k}$

Factor loading matrix.

$\varepsilon \sim N(0, \Psi)$

Specific unique noise.

$\Sigma = V V^T + \Psi$

Model covariance structure.

VARIANCE DECOMPOSITION

Factor Loadings, Communality & Uniqueness

Communality (h_i^2)

$$h_i^2 = \sum_{j=1}^{k} v_{ij}^2$$

Variance fraction of variable *xi* shared with all common factors.

Uniqueness / Specific Variance (ψ_i)

$$\psi_i = \sigma_i^2 - h_i^2$$

Remaining residual variance unique to variable *xi* alone.

Fundamental Identity: Total Variance = Communality + Specific Variance = 1.0 (for standardized data)

ALGORITHM WORKFLOW

How Factor Analysis Operates: Step-by-Step

01

Standardization

Normalize variables to zero mean and unit variance.

02

Correlation Matrix

Compute correlation matrix *R* among all variables.

03

Factor Extraction

Estimate loading matrix *V* via PCA or Maximum Likelihood.

04

Factor Rotation

Rotate axes (e.g. Varimax) for interpretable alignment.

INTERPRETABILITY OPTIMIZATION

Factor Rotation: Uncovering Simple Structure

Orthogonal Rotation (Varimax)

Maintains perpendicular 90° factor axes (uncorrelated factors).

Maximizes variance of squared loadings so each variable loads heavily onto only one factor.

Oblique Rotation (Promax)

Allows axes to correlate at non-90° angles.

Used when underlying real-world psychological or market constructs naturally interact.

WORKED EXAMPLE • FACTOR ANALYSIS

Numerical Example: Student Aptitude Test

Course	Factor 1 (vi1)	Factor 2 (vi2)	Communality (h2)	Uniqueness (ψ)
Math	0.90	0.10	0.81 + 0.01 = 0.82	0.18
Physics	0.85	0.15	0.72 + 0.02 = 0.74	0.26
English	0.10	0.88	0.01 + 0.77 = 0.78	0.22
History	0.15	0.82	0.02 + 0.67 = 0.69	0.31

Interpretation

Factor 1: Quantitative Reasoning (heavily loaded on Math & Physics).

Factor 2: Verbal Aptitude (heavily loaded on English & History).

EVALUATION • FACTOR ANALYSIS

Applications, Advantages & Limitations of FA

Key Applications

Psychometrics & IQ assessment.

Consumer satisfaction surveys.

Financial portfolio risk decomposition.

Core Advantages

Discovers actual generative causes.

Separates common variance from noise.

Enhances data interpretability.

Key Limitations

Requires linear Gaussian assumptions.

Factor rotation is non-unique.

Sensitive to sample size discrepancies.

TOPIC 2 • GEOMETRIC PROXIMITY MAPPING

Multidimensional Scaling (MDS)

Reconstructing spatial coordinate maps from pairwise object similarities and distances.

DEFINITION • MULTIDIMENSIONAL SCALING

What is Multidimensional Scaling?

MDS takes pairwise proximities (distances) between objects and places each object into a low-dimensional coordinate space.

Input: Matrix of distances or dissimilarities $D \in \mathbb{R}^{n \times n}$.

Output: Point coordinates $y_i \in \mathbb{R}^k$ (typically $k = 2$ for plotting).

Goal: Preserved Euclidean distances $||y_i - y_j|| \approx d_{ij}$.



Distance Matrix → 2D Geometric Map

DIMENSIONALITY REDUCTION ROLE

Why Perform Reduction Using MDS?

Raw Feature Absence

Operates directly on pairwise distances without raw feature vectors.

Perceptual Mapping

Visualizes subjective human judgments of taste or similarity.

Topology Preservation

Retains geometric relationships between clusters in low dimensions.

PROXIMITY FOUNDATIONS

Similarity, Dissimilarity & Distance

Similarity (s_{ij})

Higher values indicate closer, more identical objects.

$$d_{ij} = \sqrt{(s_{ii} + s_{jj} - 2s_{ij})}$$

Dissimilarity / Distance (d_{ij})

Satisfies metric properties: $d_{ij} \geq 0$, $d_{ii} = 0$, and $d_{ij} = d_{ji}$.

$$d_{ij} + d_{jk} \geq d_{ik} \text{ (Triangle Inequality)}$$

INPUT REPRESENTATION

The Proximity Matrix Structure

$$D = \begin{bmatrix} 0.0 & 3.0 & 4.0 \\ 3.0 & 0.0 & 5.0 \\ 4.0 & 5.0 & 0.0 \end{bmatrix}$$

Zero Diagonal

Distance from any point to itself is strictly zero ($d_{ii} = 0$).

Exact Symmetry

Distance from object *i* to *j* equals distance from *j* to *i*.

MATHEMATICAL DERIVATION • ALPAYDIN

Classical MDS: From Distances to Inner Products

$$B = -\tfrac{1}{2} H D^{(2)} H$$

Double-centering conversion of squared distance matrix $D^{(2)}$ into inner product matrix B

Centering Matrix H

$H = I - (1/n)\, 1\, 1^T$

Subtracts row and column means.

Spectral Recovery

$B = V \Lambda V^T \Rightarrow X = V_k \Lambda_k^{1/2}$

Yields low-dimensional coordinates X.

EXECUTION PIPELINE

Step-by-Step Classical MDS Algorithm

01

Square Distances

Form matrix $D^{(2)}$ with elements d_{ij}^2.

02

Double Centering

Compute $B = -0.5 \cdot H D^{(2)} H$ to center at the origin.

03

Eigendecompose

Extract top k positive eigenvalues and eigenvectors.

04

Extract Coords

Set coordinates $X = V_k \Lambda_k^{1/2}$.

WORKED EXAMPLE • MULTIDIMENSIONAL SCALING

Numerical Example: 3-City Map Reconstruction

Input Distance Matrix D (in km)

	City A	City B	City C
City A	0	3	4
City B	3	0	5
City C	4	5	0

Reconstructed 2D Coordinates

City A: (0.0, 0.0) [Reference Right Angle vertex]

City B: (3.0, 0.0) [Distance AB = 3.0]

City C: (0.0, 4.0) [Distance AC = 4.0]

Verification: $\sqrt{(3^2 + 4^2)} = 5.0$ (Distance BC exact!)

EVALUATION METRIC

Kruskal's Stress & Goodness of Fit

$$\text{Stress} = \sqrt{\left[\sum (d_{ij} - \|y_i - y_j\|)^2 \,/\, \sum d_{ij}^2 \right]}$$

Measures distortion between original distances and low-dimensional projected distances

< 0.05	0.05 - 0.10	0.10 - 0.20	> 0.20
Excellent Fit	Good Fit	Fair Fit	Poor Fit

EVALUATION • MULTIDIMENSIONAL SCALING

Applications, Advantages & Limitations of MDS

Applications

Market brand positioning maps.

Genetic distance visualization.

Sensory analysis of aromas and tastes.

Advantages

Requires only distance measurements.

Handles subjective similarity tables.

Visualizes complex manifolds in 2D.

Limitations

$O(n^3)$ eigendecomposition scale.

Sensitive to outlier distances.

Coordinate axes have no innate meaning.

TOPIC 3 • SUPERVISED DISCRIMINATION

Linear Discriminant Analysis (LDA)

Supervised dimensionality reduction maximizing class separation and minimizing intra-class spread.

CONCEPT OVERVIEW • LINEAR DISCRIMINANT ANALYSIS

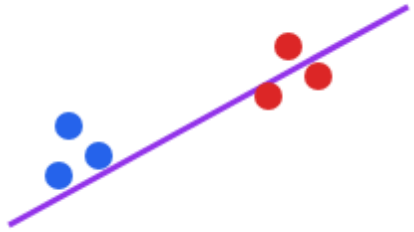
What is Linear Discriminant Analysis?

LDA finds a linear combination of features that best separates two or more classes of objects.

Supervised Learning: Uses class labels during dimensionality reduction.

Projection Axis *w*: Projects high-dimensional data onto a 1D line or lower subspace.

Dual Objective: Far-apart class means + tight within-class clusters.



Optimal Separation Projection Line *w*

METHOD COMPARISON

Supervised (LDA) vs. Unsupervised (PCA)

Unsupervised (PCA / FA)

Ignores class labels completely.

Finds directions of overall maximum variance.

Risk: May project distinct classes directly onto each other!

Supervised (LDA)

Explicitly uses target class labels $r \in \{1, ..., K\}$.

Finds directions that maximize between-class spread.

Ensures optimal boundary placement for classifiers.

MATHEMATICAL FORMULATION

Within-Class Scatter Matrix (S_W)

$$S_W = \sum_{i=1}^{C} \sum_{x \in C_i} (x - m_i)(x - m_i)^T$$

Measures total dispersion and spread of samples around their own class centroid

Two-Class Case

$S_W = S_1 + S_2$, sum of individual scatter matrices.

Optimization Goal

We want $w^T S_W w$ to be as small as possible (tight clusters).

MATHEMATICAL FORMULATION

Between-Class Scatter Matrix (S_B)

$$S_B = (m_1 - m_2)(m_1 - m_2)^T$$

Outer product representing the squared distance between the two class means

Geometric Meaning

Measures how far apart the centers of the classes lie in feature space.

Optimization Goal

We want $w^T S_B w$ to be as large as possible (maximum separation).

OPTIMIZATION OBJECTIVE

Fisher's Linear Discriminant Criterion

$$J(w) = (w^T S_B w) / (w^T S_W w)$$

The Rayleigh quotient balancing between-class variance over within-class variance

Numerator

Maximizes separation between projected class centroids.

Denominator

Minimizes spread/variance within each projected class.

CLOSED-FORM DERIVATION

Mathematical Solution for the Projection Vector

Optimal Vector w

$$w = S_W^{-1} (m_1 - m_2)$$

Derived by taking the derivative of $J(w)$ with respect to w and equating to zero.

Decision Threshold w_0

$$w_0 = -\tfrac{1}{2} w^T (m_1 + m_2)$$

Classify sample x into Class 1 if $w^T x + w_0 > 0$, else into Class 2.

EXECUTION PIPELINE

Step-by-Step LDA Algorithm

01

Compute Means

Calculate mean vectors m_1 and m_2.

02

Compute Sw

Calculate scatter matrices $S_1 + S_2$.

03

Invert Sw

Compute matrix inverse S_W^{-1}.

04

Calculate w

Multiply $w = S_W^{-1}(m_1 - m_2)$.

05

Project & Test

Project samples: $y = w^T x$.

WORKED EXAMPLE • LINEAR DISCRIMINANT ANALYSIS

Numerical Example: Two-Class 2D Projection

Training Points

Class 1: (1, 2) and (2, 3) → Mean $m_1 = [1.5, 2.5]^T$

Class 2: (6, 5) and (7, 7) → Mean $m_2 = [6.5, 6.0]^T$

Difference of Means

$m_1 - m_2 = [-5.0, -3.5]^T$

Scatter & Projection

$S_W = \text{diag}(1.0, 2.5) \Rightarrow S_W^{-1} = \text{diag}(1.0, 0.4)$

$w = [-5.0 \times 1.0, -3.5 \times 0.4]^T = [-5.0, -1.4]^T$

Result: Perfect class separation along the 1D projection axis!

EXTENSION • ALPAYDIN FORMULATION

Multi-Class LDA (C > 2 Classes)

Maximum Projection Subspace

$$k \leq C - 1$$

Because C class centroids span at most a $(C - 1)$-dimensional hyperplane, $\text{rank}(S_B) \leq C - 1$.

Generalized Eigenvalue Solution

$$S_{W}^{-1} S_B W = W \Lambda$$

Columns of W are eigenvectors corresponding to the largest eigenvalues of $S_{W}^{-1} S_B$.

EVALUATION • LINEAR DISCRIMINANT ANALYSIS

Applications, Advantages & Limitations of LDA

Applications

Face Recognition (Fisherfaces).

Medical disease diagnostics.

Credit score risk classification.

Advantages

Supervised class separation.

Closed-form mathematical solution.

Low runtime computation cost.

Limitations

Singular Sw when $N < d$.

Assumes identical class covariance.

Fails on non-linear boundaries.

COMPARATIVE SYNTHESIS

Comparison: Factor Analysis vs. MDS vs. LDA

Feature / Property	Factor Analysis (FA)	Multidimensional Scaling	Linear Discriminant (LDA)
Supervision	Unsupervised	Unsupervised	**Supervised (Labels used)**
Primary Input	Covariance Matrix Σ	Pairwise Distance Matrix *D*	Feature Vectors *X* & Labels *Y*
Core Objective	Uncover latent generative causes	Preserve pairwise distances in 2D	Maximize class separability
Key Math Tool	Σ = V VT + Ψ	Double Centering *B = -0.5 H D2 H*	Fisher Criterion *SW-1SB*
Max Dimensions	User selected (*k* < *d*)	Typically *k = 2* or *3*	**At most *C - 1***

REVIEW & SUMMARY

Unit IV Key Conceptual Takeaways

Factor Analysis

Generative model that groups observed correlated attributes under common latent factors.

Multidimensional Scaling

Geometric embedding reconstructing spatial coordinate maps directly from proximity matrices.

Linear Discriminant

Supervised projection maximizing between-class scatter over within-class scatter.

EXAMINATION PREPARATION • BCA 5TH SEMESTER

High-Frequency University Exam Questions

Q1. Define latent factors and factor loadings. State the FA model formula.

Q2. Distinguish between communality and specific variance with an equation.

Q3. Explain the purpose of Varimax rotation in Factor Analysis.

Q4. What is a distance matrix in MDS? Explain double centering.

Q5. Define Kruskal's Stress and state the benchmark criteria for good fit.

Q6. Contrast supervised LDA with unsupervised PCA.

Q7. Derive Fisher's linear discriminant criterion $J(w)$.

Q8. Why is the maximum number of dimensions in LDA limited to $C - 1$?
